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Master Thesis

**“Using information theory to identify process controls in
Global Water Models”**

Author:	Maximilià-Manuel Serra Lasierra
Study program:	M.Sc. Data Science
Matriculation number:	815109
First supervisor:	Prof. Dr. Thorsten Wagener
Second supervisor:	Prof. Dr. Marina Höhne

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A FEW PRELIMINARY WORDS

This thesis has been written following the template of the “Journal of Geophysical Research: Machine Learning and Computation”, as it was initially agreed between the first supervisor and the author. Therefore, the pages that follow are formatted as the manuscript one would submit to the journal. The large blank gaps, line numbering and other formatting are due to the template being used, I apologise for these minor inconveniences.

All the code, scripts and notebooks used throughout this thesis, are available in the GitHub repository: https://github.com/maxserra/master_thesis.

The data necessary to reproduce the results is also available in the GitHub repository. Refer to the documentation in the repository for instructions.

Using information theory to identify process controls in Global Water Models

Maximilià Serra Lasierra¹

¹University of Potsdam, Germany

Key Points:

- The Maximal Information Coefficient (MIC) enables agnostic understanding of variable interactions in complex datasets.
- Application to Global Water Models enables efficient identification of complex hydrological relationships hidden for traditional measures of correlation.
- In-depth study of relationships detected using MIC reveal insights in the process modeling assumptions and constructs of Global Water Models.

Abstract

Global Water Models (GWMs) are key for understanding and projecting global hydrological fluxes under changing climate conditions, yet their outputs often diverge, limiting their utility for robust decision-making. Uncovering and identifying the relationships embedded in these complex simulation datasets requires measures of dependence that can capture a wide range of functional behaviors. In this study, we apply the Maximal Information Coefficient (MIC), an information theory based, non-parametric measure of dependence, to systematically explore and characterize the input-output relationships in the Community Water Model (CWatM) forced with ISIMIP3a observed climate data. Our results demonstrate that MIC not only recovers expected hydrological controls but also reveals previously unnoticed functional relationships that Pearson and Spearman correlation coefficients would have overlooked. To further explain the complex dependencies found in the data, we use a Random Forest classifier and permutation feature importance to isolate key explanatory factors from the model’s internal structure and domain-specific factors. This integrated, information theory based approach provides a systematic methodology to improve model diagnostic capabilities, informing model development, guiding targeted research directions, and ultimately strengthening the credibility and interpretability of large-scale hydrological simulations.

Plain Language Summary

Global Water Models are global-scale computer simulations used to predict how water moves around the world through rainfall, rivers and groundwater fluxes. These models are increasingly important as the consequences of climate change are altering the world. However, these models often produce conflicting results, making it challenging to decide which predictions to trust when planning for things like droughts, floods, and water use. In this study, we analysed one such Global Water Model and asked a simple question: How can we uncover the underlying connections that shape the relationship between the inputs and outputs? Traditional methods focus on straightforward relationships, like whether more rain leads directly to more river flow. But nature, and by extension these models, are more complex than that. To tackle this, we use a tool from the field of information theory, the Maximal Information Coefficient. This can be thought of as a dependency detector that can spot complex patterns and subtle connections better than other methods miss. In applying this method, we discovered not only the expected relationships (such as high temperatures leading to more potential evaporation) but also new and unexpected patterns that standard approaches would have overlooked. We then successfully used a machine learning technique to figure out which factors in the model create these hidden patterns, uncovering where the model behaved unexpectedly and where it might need refining. The proposed approach, provides relevant and insightful tools to assist with the evaluation of Global Water Models. Better evaluated models will yield more reliable predictions, which ultimately help everyone, from local communities to international policymakers, in making better decisions about water in a changing world.

1 Introduction

Gaining a better understanding of the global water cycle is increasingly important, as water scarcity and extreme weather events are happening more frequently and strongly due to the increasing impact of climate change (Pokhrel et al., 2021). The unforeseeable effects that a changing climate can have on the global water cycle, together with anthropogenic factors such as agricultural (Wada et al., 2012) and urban expansion (Ozturk et al., 2022), stress the importance of deepening our understanding of the hydrological processes that govern water availability (He et al., 2017). Better insights into the water cycle at a global scale, are demanded of current models, which aspire to provide realistic and robust projections of future scenarios under changing climatic conditions (Wada et al., 2016). Projections of hydrological variables such as streamflow, groundwater recharge and terrestrial water storage are the basis upon which, for example, the Intergovernmental Panel on Climate Change bases its assessment reports and recommendations for policy makers.

Global Water Models (GWMs) are used to generate these projections, but their outputs are subject to substantial uncertainty and frequently diverge from one another in both historical and future scenario simulations (Gnann et al., 2023). These discrepancies reduce credibility and make effective policy support difficult (Krysanova et al., 2018). GWMs represent the current understanding the scientific community has of the water cycle (Wagener et al., 2021), they include the relevant hydrological processes from precipitation to runoff to simulate the water cycle at a global scale. However, these models are commonly built following a bottom-up approach, focusing on accurately simulating processes at the grid-cell level rather than ensuring coherent global behavior. As a result, their global-scale dynamics are only studied once the simulation results are available and these might not be as intended. Moreover, projections from different models are often divergent, producing large uncertainties that render model outputs less informative to decision makers (Krysanova et al., 2018). This considerable disagreement among models, which is not consistent across the globe (Gnann et al., 2023), highlights the need for a deeper understanding of the sources of these discrepancies and for improved methods to evaluate and compare GWMs.

To assist with the improvement of these models and to continue to provide valuable information to policy makers, functional relationships have been proposed as a building block of a new evaluation strategy that can provide insights into how GWMs represent the water cycle (Wagener et al., 2021). Functional relationships describe how a (hydrological) system processes inputs to produce outputs, thereby showing how two or more variables are connected and influence each other (Gnann et al., 2023). By examining how models capture these relationships between climatic forcings and hydrological responses at a global scale, we can gain a better understanding of the modeling assumptions and identify areas for improvement (Wagener et al., 2022). Similarly to the objective of the present study, methods such as the Mean Decrease in Impurity of Random Forests have been proposed as a diagnostic tools to understand the relative importance of different input variables in model simulations (Brêda et al., 2024). However, the functional relationships between hydrological variables are inherently complex and oftentimes cannot be easily described by simple functions. This complexity is unlikely to be fully captured by traditional measures of dependence, such as Spearman’s rank correlation (Spearman, 1904), which may fail to account for the non-linear interactions present in these systems. The work of Brêda et al. (2024) leverages a surrogate model ensemble to approximate this complexity, but as the authors acknowledge, the generality of their findings is limited to that of the surrogate model, which limits the extent of the conclusions that can be drawn from such analysis.

Information theory inspired measures of dependence, such as the maximal information coefficient (MIC), offer a promising alternative to traditional measures for detecting relationships in large and complex datasets (Reshef et al., 2011). Mutual infor-

mation (Shannon, 1948), upon which the MIC is defined, is designed to measure the shared information between two variables, effectively quantifying the strength of the relationship between variables without assuming a specific functional form. Therefore, a measure such as the MIC should be able to capture more and different relationships that might have previously gone unnoticed. Relationships detected on the output data of the model under evaluation might be the product of a complex underlying process, and direct interpretation might not be possible. In such cases, a surrogate model can be used to guide the search for an explanation (Brêda et al., 2024).

In this study, we will leverage information theory to explore the simulation results of one widely used Global Water Model, the Community Water Model (Burek et al., 2020). The simulation results we will use were produced as part of the Inter-Sectoral Impact Model Intercomparison Project (ISIMIP), specifically we will use the data from the ISIMIP3a (Frieler et al., 2024) with observed climate forcings. Our aim is to identify the processes that control the model outputs in certain domains with the goal of evaluating if and how much this representation of the terrestrial water cycle aligns with the perceptual model shared among the hydrological community (Wagener et al., 2021). In observing unintended model behavior, we will provide an example of using a Random Forest model to understand the unexpected patterns and find a plausible explanation.

2 Methods

As datasets of tens and hundreds of variables are becoming more and more common across multiple disciplines, the challenge of exploring them effectively is growing. In the scope of this study, the objective of exploring such datasets is to identify strong and undiscovered relationships between variable pairs. As it becomes unfeasible to examine all variable pairs ad-hoc, a systematic procedure is required. The approach we follow in this study is to compute a set of measures of dependence for each pair and focus the attention on those variable pairs with the highest scores. In section 2.1 we look into different measures of dependence and their suitability for this task.

Once we find such strong and undiscovered relationships it might not be easy to explain them at naked eye. In this study we will use Random Forest classifiers, to search for an explanation to such relationships. By using the permutation feature importance of the fitted classifier, we can guide our exploratory efforts towards the variables that are most relevant, in hope to find satisfactory explanations to the found relationships. In section 2.2 we explain these models and how we use them in this study.

2.1 Measures of dependence

The most widely known and used measures of dependence are the Pearson and Spearman correlation coefficients. Pearson correlation is a parametric measure that measures linear relationships (Pearson & Galton, 1997), Spearman correlation measures rank correlation (monotonicity) (Spearman, 1904). Simple yet highly informative, these measures are great to identify specific types of relationships, linear and monotonous ones, respectively. However, they are not designed to, and therefore fail at identifying more complex relationships such as polynomials of higher order and periodic functions.

In the face of these challenges, Reshef et al. (2011) propose the maximal information coefficient (MIC), as a measure of dependence based on information theory. The MIC is based on mutual information, a non-parametric and general measure of how much information two variables share about each other. The MIC is therefore capable of detecting relationships regardless of their specific functional form, whether they are linear, monotonous, or exhibit more complex patterns such as associations that may not be described by a function.

Generally, the task of exploring a large number of variable pairs by means of a measure of dependence, requires that this measure satisfies two properties: generality and equitability (Reshef et al., 2011). Generality refers to the detection of any type of relationship, and equitability refers to the assignment of similar scores to similarly noisy relationships. These two properties, while conceptually simple, are highly important since they provide a benchmark to assess the suitability of different measures of dependence (SOM of (Reshef et al., 2011), (Reshef et al., 2013), (Kinney & Atwal, 2014) and (Reshef et al., 2014)).

2.1.1 Maximal information coefficient (MIC)

The MIC "is based on the idea that if a relationship exists between two variables, then a grid can be drawn on the scatterplot of the two variables that partitions the data to encapsulate that relationship", as described by the authors in Reshef et al. (2011). The algorithm computes the mutual information for all possible grids over the sample space of the two variables, up to a certain limit, then it normalizes the values to account for the grid size and determines the maximum achievable value as the MIC. In what follows, we introduce the foundational concepts upon which this algorithm is funded.

Mutual information was initially introduced and studied in the field of signal processing (Shannon, 1948) and is based on the concept of entropy. Entropy (represented

by H), in the domain of information theory, is a measure of how much information a given variable contains. This is defined as the expected value of the information content of a variable and can be understood as how much informational value an observation of that variable has. Intuitively, the more values a variable might take, the more information we obtain by observing an occurrence of it, and if a variable always takes one single value, there is no informational value in observing it.

Below, the equations of entropy $H(X)$, joint entropy $H(X, Y)$ and mutual information $I(X; Y)$ are formulated as functions of the probability density functions (*pdf*) $p_x(X)$ and the joint *pdf* $p_{x,y}(X, Y)$. The probability density function is a function over the sample space of a given variable (or pair of variables) which takes values between 0 and 1 and equals to the probability of occurrence of each event (or pair of events).

$$H(X) = \mathbb{E}[-\log(p_x(X))] \quad (1)$$

$$H(X, Y) = \mathbb{E}[-\log(p_{x,y}(X, Y))] \quad (2)$$

$$I(X; Y) = H(X) + H(Y) - H(X, Y) \quad (3)$$

Mutual information can be expressed in terms of entropy by equation 3. This formulation highlights the symmetry of mutual information (X and Y can be exchanged). This property, which is also shared with Pearson and Spearman correlation coefficients, is notably relevant for any candidate measure of dependence as it emphasizes the focus on the association strength and not the directionality of the relationship. Moreover, from equation 3 it can be seen that mutual information does not include any assumption on the function that might relate X with Y . In this sense, it has the potential to fulfill the generality property (Reshef et al., 2011) discussed previously, since H depends only on the probability density functions of the variables in a completely shape agnostic way.

The empirical estimation of mutual information is, therefore, highly dependent on the empirical *pdf* of the data. If a grid is chosen to be too coarse, it might not be able to capture the complexity in the data, but it is chosen to be too fine, it will likely find a relationship when there is none. This is why the MIC algorithm explores multiple grid sizes and normalizes the computed mutual information by a factor which is closely related to the size of the grid and the maximum attainable mutual information. In doing so, this normalized value of the mutual information ensures a fair comparison between different grids and provides a value between 0 and 1. Finally, the MIC value for a pair of variables is defined to be the maximum over all the explored grid sizes.

The MIC algorithm, illustrated in figure 1, can be broadly summarized in three main steps. In step A, various grid sizes and splitting locations are evaluated and stored in what the authors refer to as the characteristic matrix. This matrix, as represented in B, contains the optimal splits for a given number of rows and columns, with values representing the maximal attainable normalized mutual information for each grid size. Finally, as shown in C, the maximum value from the characteristic matrix is selected.

Additionally, the characteristic matrix computed in the MIC algorithm can be used to derive further statistics that serve as an indicator of certain relationship types. In Reshef et al. (2011) the authors define three such statistics; the maximum asymmetry score (MAS), the maximum edge value (MEV) and the minimum cell number (MCN). The MAS is portrayed as a measure capable of detecting departure from monotonicity, which in combination with a high value of MIC could be used as an indicator of periodic relationships or other interesting functional dependencies. We refer the interested reader to the original paper for more information on the other two derived statistics (SOM of (Reshef et al., 2011)).

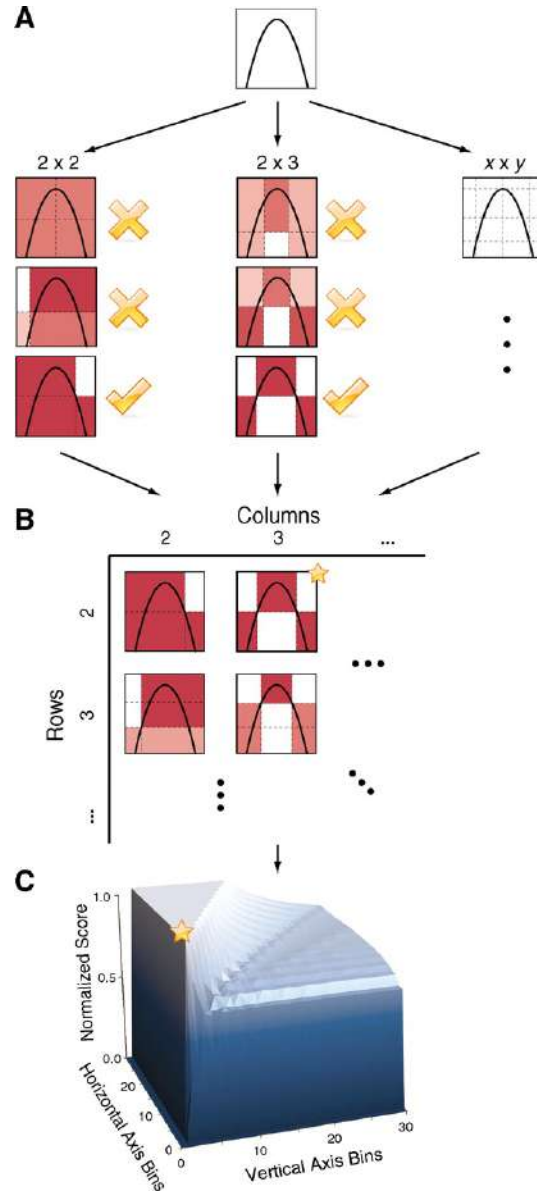


Figure 1: MIC algorithm diagram. Figure from the original paper by Reshef et al. (2011).

Therefore, the MIC is a measure of dependence that satisfies the desirable properties of generality and equitability, enabling the exploration of large data sets in an effective way. It leverages the generality of information theory and mutual information, and implements a sensible and clever splitting and normalization strategy that accounts for the risks of binning to provide an equitable normalized value. Moreover, it provides a set of derived statistics that have the potential to be of interest when identifying functional relationships.

2.2 Random forests and permutation feature importance

Random forests are ensemble models that combine the predictions of multiple decision trees to provide more robust and general predictions (Breiman, 2001). Decision trees are machine learning models that recursively split the space to create a structure of hierarchical decisions aimed at classifying or predicting a target variable. The splits aim to optimize metrics such as entropy or Gini impurity, ultimately forming a flowchart-like model that is easily interpretable (Breiman et al., 2017). However, a single decision tree can be overly sensitive to small variations in the data, potentially leading to unstable predictions and overfitting.

Random forests address the limitation of single decision trees by training an ensemble them, each on a random subset of the data and its features. By combining a large collection of diverse trees, a random forest provides improved predictive accuracy and robustness relative to a single decision tree. This ensemble approach leverages the idea that while individual trees may fail to detect certain patterns in the data, this can be circumvented by a collective decision, which tends to be more stable and generalizable (Breiman, 2001). Intuitively, each tree in the ensemble has the potential to learn different characteristics of the data, and by aggregating the result of multiple of these trees, the result will be better.

An interesting and useful property of random forests, beyond their improved predictive performance, lies in their ability to quantify the relative importance of input variables (De la Fuente et al., 2023). Traditionally, tree-based models measure importance through the mean decrease in impurity (MDI), which quantifies how much each feature contributes to reducing uncertainty or impurity within the forest (used in Brêda et al. (2024)). However, this method has notable limitations, such as bias towards features with higher cardinality and the potential to overestimate the importance of variables in overfitted models. These limitations can obscure the true relevance of features in complex datasets, particularly in scenarios where features vary in their scales and/or distributions.

Permutation feature importance provides a more robust and model-agnostic alternative for assessing the relevance of predictors (Breiman, 2001). This method involves randomly shuffling the values of a single feature and observing the resulting degradation in the model’s performance. By breaking the relationship between the feature and the target variable, permutation importance is a measure of how much the model relies on each feature to make correct predictions.

Another advantage of permutation feature importance is its flexibility in leveraging various performance metrics, such as accuracy, area under the ROC curve (Hand & Till, 2001), or mean squared error, to evaluate feature contributions. This versatility ensures that the importance rankings align closely with the specific objectives and evaluation criteria of the analysis. By providing unbiased and performance-driven insights into the contribution of each predictor, permutation feature importance offers a more reliable tool for guiding model interpretation and the subsequent understanding of the relationships between inputs and outputs.

3 Data (CWatM and ISIMIP3a)

3.1 The Community Water Model

For this study we selected the Community Water Model (CWatM), which stands out as one of the most comprehensive Global Water Models available (Burek et al., 2020). The CWatM model is designed to provide insights on the socioeconomic changes derived from future water availability and demand scenarios. To this end, a variety of relevant processes are included into the model, ranging from agricultural, industrial and domestic water demand to soil infiltration and groundwater flows. Together with accurate representation of the hydrological processes and energy balances, the CWatM simulations lead to non-trivial relationships in the data.

The CWatM model inputs data which is either static or time varying. Static data includes landsurface characteristics such as elevation, river drainage and channel maps, soil and its hydraulic properties, groundwater, lakes and reservoirs. Time varying inputs include landcover (forest, grasslands, irrigated areas, paddies and sealed/impermeable surfaces) which change yearly, and meteorological variables which are used with a daily resolution. These meteorological variables are commonly referred to as forcings and characterize the specific simulation scenario.

3.2 ISIMIP3a protocol

For this study we used the data generated in the scope of the Inter-Sectoral Impact Model Intercomparison Project (ISIMIP), this project provides a framework for simulating and assessing climate impact models. Specifically, we use the latest available run, the ISIMIP3a. This protocol covers historical simulations where models are forced with both observed and de-trended climate forcings and with observed socioeconomic data to provide a consistent basis for model evaluation and impact attribution (Frieler et al., 2024). We focused this analysis to the observed climate scenario and with the GSWP3-W5E5 climate forcing dataset.

3.3 Data preparation and processing

The data preparation and processing done in this study is depicted in figure 4. We first generated static maps of all the variables by computing the 30-year (climatological) averages (1975-2004) for all the variables. The forcings and model outputs are openly available on the ISIMIP data repository and were obtained from the website. On the other hand, the model inputs concerning landcover were obtained directly from the group running the CWatM model to ensure that we worked with the exact same datasets inputted to the model run and to reduce the risk of inconsistencies. Fluxes were converted to mm/year for consistency across variables. Data was then masked to land cover (GWMs don't simulate the water fluxes over the ocean).

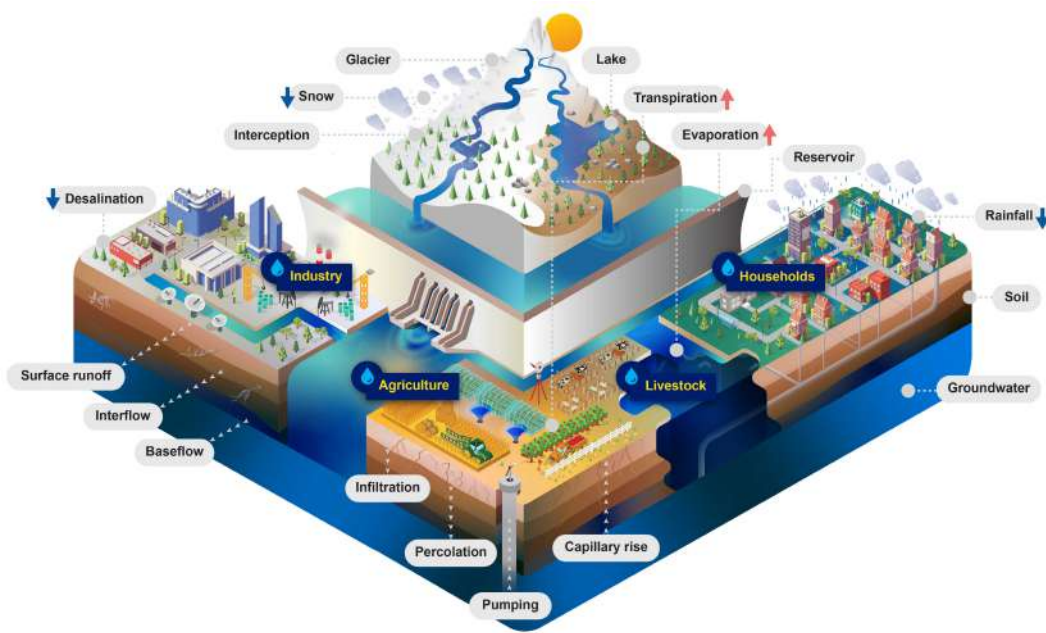


Figure 2: Community Water Model (CWatM) diagram. Figure from the official model documentation by Burek et al. (2020).

Ground Water	Land Cover	Inputs				Outputs
		Land Surface	Routing	Soil	Forcings	
kSatAquifer	interceptcap_forest	Global_CloneMap_30min	area	alpha1	hurs	evap-total
porosity	interceptcap_grassland	albedoLand	cellarea30min	alpha2	huss	potevap
recessionCoeff	interceptcap_irrNonPaddy	albedoWater	chanbnkf	alpha3	pr	twos
	interceptcap_irrPaddy	domwc	chanbw	cropgrpn_o	ps	qtot
	kc_forest	domww	changrad	firstStorDepth	rlds	qr
	kc_grassland	dzRel0000	chanleng	ksat1	rsds	
	kc_irrNonPaddy	dzRel0001	chanman	ksat2	sfcwind	
	kc_irrPaddy	dzRel0005	discharge	ksat3	tas	
	maxRootDepth_forest	dzRel0010	lakeresArea	lambda1	tasmax	
	maxRootDepth_grassland	dzRel0020	lakeresDis	lambda2	tasmin	
	maxRootDepth_irrNonPaddy	dzRel0030	lakeresID	lambda3		
	maxRootDepth_irrPaddy	dzRel0040	lakeresType	percolationImp		
	minSoilDepthFrac_forest	dzRel0050	lakeresVolRes	secondStorDepth		
	minSoilDepthFrac_grassland	dzRel0060	lakeresVolTotal	thetar1		
	minSoilDepthFrac_irrNonPaddy	dzRel0070	lakeresYear	thetar2		
	minSoilDepthFrac_irrPaddy	dzRel0080	ldd	thetar3		
	rootFraction1_forest	dzRel0090	ups	thetas1		
	rootFraction1_grassland	dzRel0100	watershedarea	thetas2		
	rootFraction1_irrNonPaddy	efficiency		thetas3		
	rootFraction1_irrPaddy	elvstd				
	rootFraction2_forest	fracforest				
	rootFraction2_grassland	fracgrassland				
	rootFraction2_irrNonPaddy	fracirrNonPaddy				
	rootFraction2_irrPaddy	fracirrPaddy				
		fracsealed				
		fracwater				
		indwc				
		indww				
		irrigationArea				
		livestockConsumption				
		tanslope				

Figure 3: List of the CWatM inputs and outputs considered in this study. Note that only the variables used in this study are included, the model produces more outputs that are not listed above. The input variables are grouped and colored according to their domain. We present the variable names used in the model and in the ISIMIP repository, we refer the interested reader to the corresponding sources for the variable descriptions.

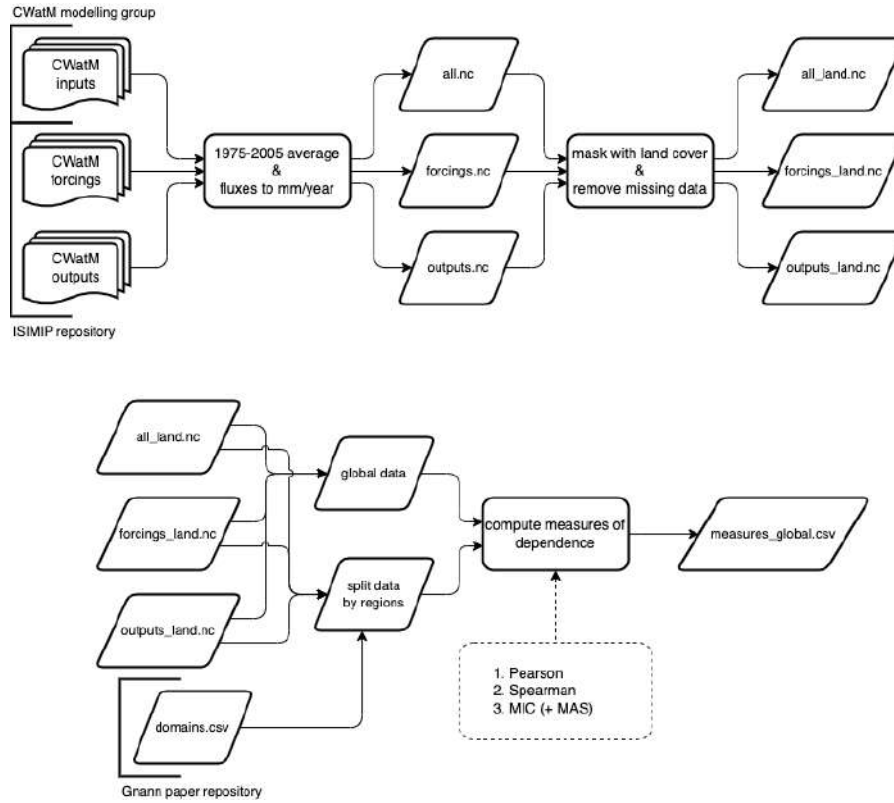


Figure 4: Schematic representation of the data processing flow used in this study. The computation of the measures of dependence computation is also shown. All the code relevant in this diagram is openly available in the thesis GitHub repository.

4 Results

4.1 Evaluating the Utility of the Maximal information coefficient

To determine whether the maximal information coefficient (MIC) provides different information than the Pearson and Spearman correlation coefficients do, we compare the three measures of dependence by listing the top six inputs that are most strongly related to five outputs of the CWatM model. In figure 5 we show how each measure gives different ranking to each input, and it can be seen how some variables are only considered relevant for one measure. The differences in ranking of certain variables, and the appearance of variables indicates that the different measures are capturing different relationships. For example, it can be seen that for the output variable potential evapotranspiration (potevap), the MIC keeps the maximum temperature (tasmax) as the most strongly controlling variable, while giving water albedo (albedoWater) and river channel length (chanleng) a higher ranking than Pearson and Spearman do (scatterplots of these relationships shown in figure 6). Another interesting case is that of ground water recharge (qr), where precipitation (pr) is ranked the highest for both Pearson and Spearman, but it goes to fourth place for MIC and percolation impediment (percolationImp) appears as the most strongly controlling variable (scatterplots in figure 7).

We acknowledge that observing different measures of dependence providing different ranking should not come as a surprise, this is somewhat to be expected. What we consider to be significant here is that the variables that gain importance make sense from both a physical and modeling perspective. Taking a closer look at the case of potential evapotranspiration (potevap), three of the top six variables for all three measures are temperature related, and this is to be expected since temperature strongly drives the potential for both evaporation and transpiration. However, we see that water albedo (albedoWater) is only appearing for the MIC, this is reasonable since the capacity for a water body to absorb radiation should be closely linked to its capacity to change phase. In figure 6b it can be seen that the values of Pearson and Spearman are not low, in fact they are higher than the MIC, but from the ranking in figure 5 we know that other variables have higher values.

In the case of ground water recharge (qr), we observe a different situation (see figure 7); precipitation (pr) and percolation impediment (percolationImp) show decreasing and increasing values of the dependence measures, respectively. This matches the trend observed in their ranking values in 5.

From a methodological perspective, we would like to mention that while the value of the measures themselves are important, we consider the ranking (as presented in figure 5 and A2) to be an insightful representation of this results. When looking at the ranking rather than the values of the measures, one can easily see which variables each measure is prioritizing, and this can help focus the analysis to those.

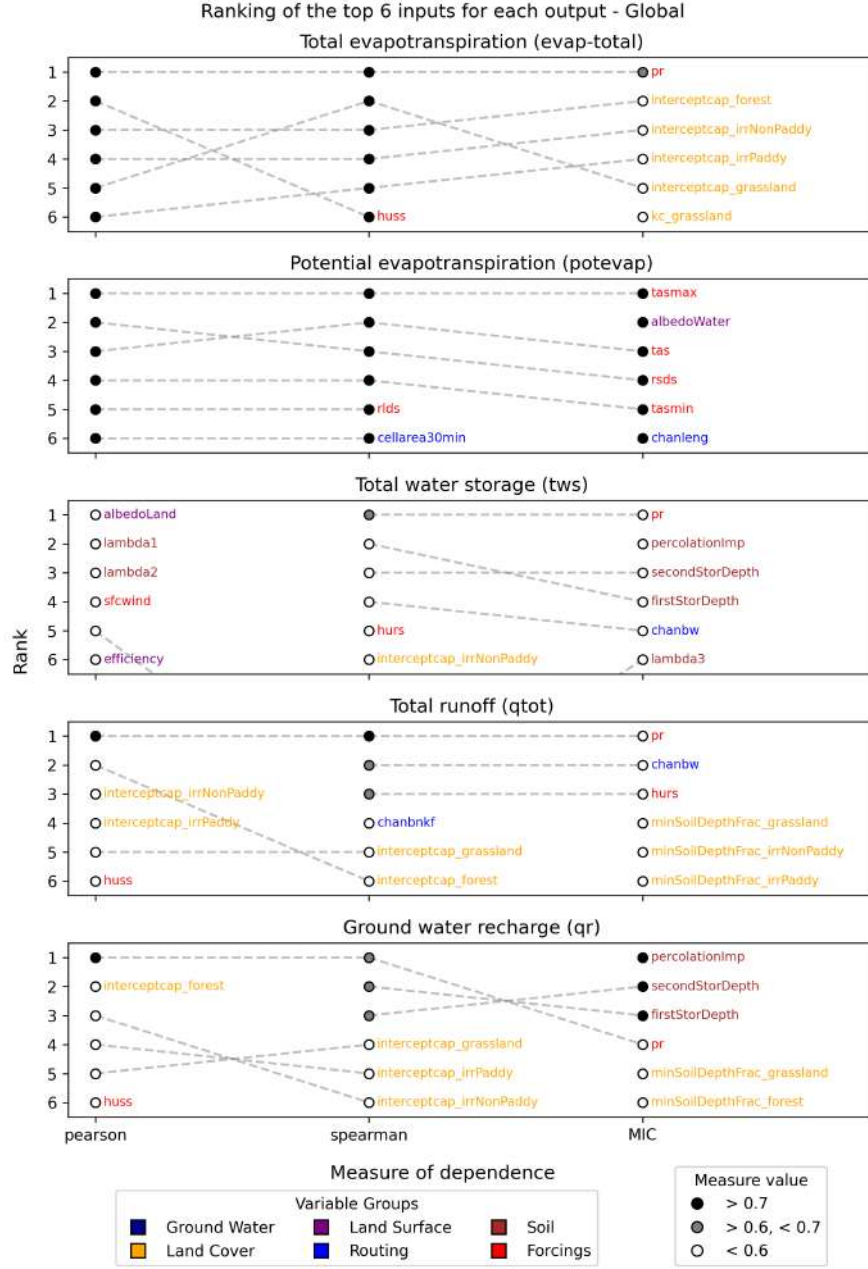


Figure 5: Ranking of the top 6 most controlling inputs for 5 selected outputs according to Pearson, Spearman and MIC. These measures are computed on the global dataset. The input variables are grouped and colored according to their domain. To indicated the actual value of the measures of dependence, the dots are colored black for value above 0.7, gray for values between 0.6 and 0.7 and white for those below 0.6. A table with numerical values is provided in A1.

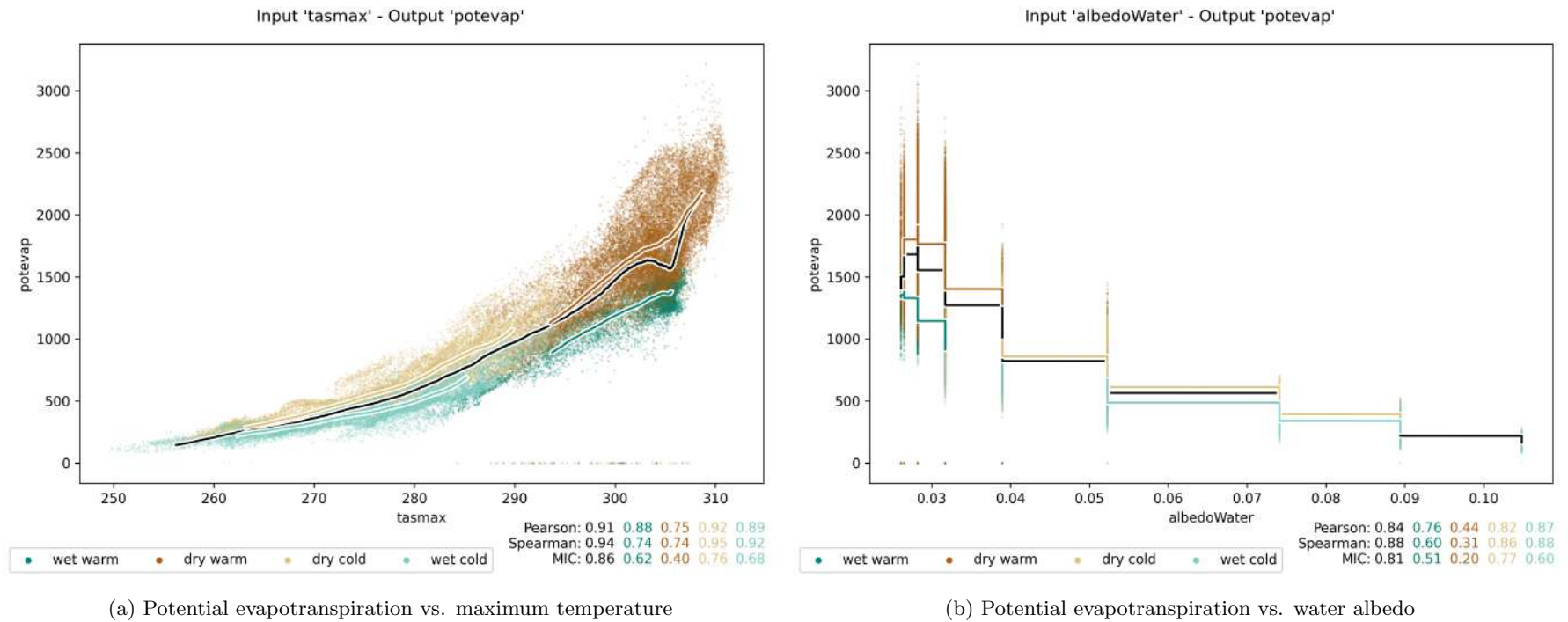


Figure 6: Scatterplots of potential evapotranspiration against maximum temperature and water albedo. Data is colored according to the climate regions defined by Gnan (see Appendix B), black corresponds to the joint global dataset. The lines show the average relationship per region. In the bottom right corner, the values of the three measures of dependence are shown.

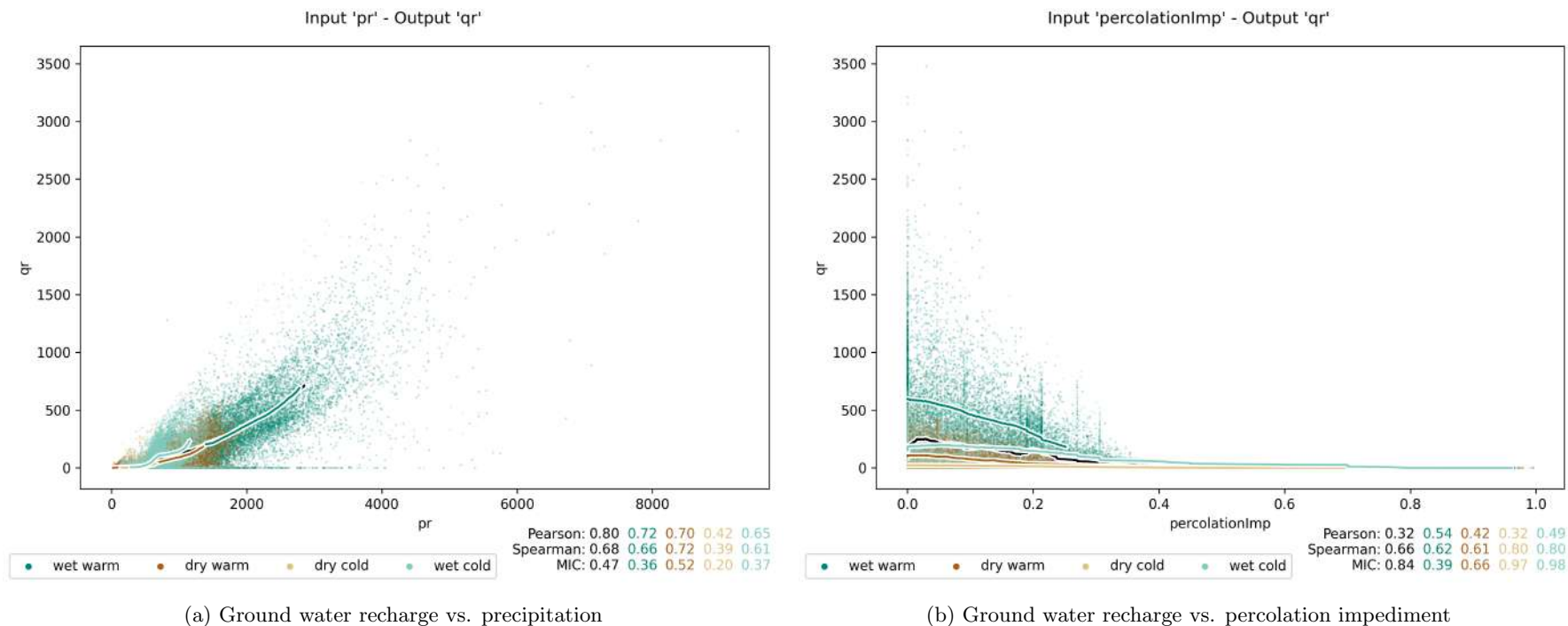


Figure 7: Scatterplots of ground water recharge against precipitation and percolation impediment. Data is colored according to the climate regions defined by Gnnann (see Appendix B), black corresponds to the joint global dataset. The lines show the average relationship per region. In the bottom right corner, the values of the three measures of dependence are shown.

4.2 Distinguishing Relationship Types

Having determined that MIC is a valuable measure of dependence that is able to detect relationships that both Pearson and Spearman might not have been able to detect, we set off to determine and characterize relationship types that can be identified with these set of measures. In figure 8 we show the metrics profiles of three relationship types that can be clearly determined based on their value of Pearson, Spearman, MIC and MAS. With this classification we show how different functional relationships can be identified based on the values of the computed metrics. This enablement is specially relevant for cases where the number of variable pairs to explore is large, since this profiles can guide the search for specific types of relationships.

Moreover, we would like to emphasize that these three types of relationships would have gone unnoticed without the MIC, as shown with the 0.7 threshold in 8. While the 0.7 value is somewhat arbitrary, it is common practice in the community to set such a threshold to determine the significance of Pearson and Spearman correlation. Here we apply this threshold to the MIC too, but we do so only for illustrative purposes and not to claim this as a new hard-set threshold.

The relationship types identified and reported in figure 8 can be understood by examining the values of each measure of dependence. For instance, simple relationships are characterized by high Pearson, Spearman and MIC values, indicating that these relationships are simultaneously linear, monotonic, and strong. In contrast, threshold-like relationships display a low Pearson value, which reflects the non-linear trend shown, but maintain a higher Spearman coefficient, suggesting a generally monotonic trend (either increasing or decreasing). Their high MIC values confirm the strong overall dependence despite the non-linear pattern. Finally, complex relationships are characterized by high MIC values paired with low Pearson and Spearman correlations, implying that although the variables are strongly related, their relationship is neither linear nor monotonic. In these complex cases, the MAS measure is particularly valuable, as its high values highlight the departure from simple, monotonic patterns.

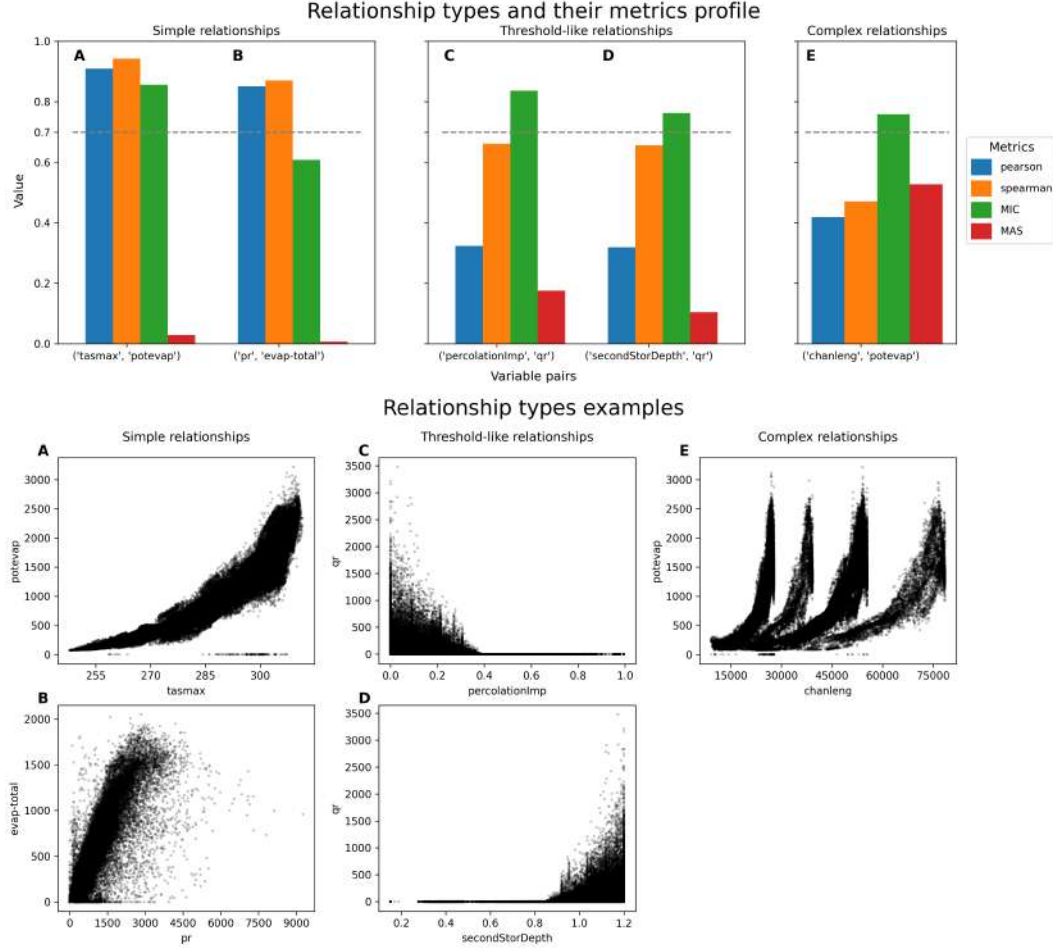


Figure 8: Relationship types that can be recognized based on the values and profile of the computed metrics. The bar plots show the metrics profile of the three different relationship type, the letters link the barplots with the respective scatterplots below, which exemplify the identifiable relationships. We present the three relationship types or patterns we have identified in the CWatM dataset, application of this method to further datasets might uncover other relevant trends. The numerical values of each metric can be found in the table A1.

4.3 Explaining Complex Patterns

The patterns observed in the scatter plot of channel length and potential evapotranspiration (see figure 8 E) expose a strange and strong relation between these two variables. From a hydrological perspective it is reasonable to observe that locations with greater channel length yield more potential evapotranspiration, however one does not expect to observe four clearly distinct functional relationships.

To understand these four trends, we used a random forest classifier with manually labeled classes matching the four functions that are visible with the naked eye (see Appendix E). The trained classifier model performed extremely well (0.984 ROC area under the curve) and by analyzing the permutation feature importance we could determine the variables the model considered most relevant to determine these four regions (for details on the Random Forest evaluation see Appendix E, for the permutation feature importances, see figure E2). The two variables that stand out as the most important are the upstream water area (ups) and the local drain direction (ldd), which are both river routing variables.

In figure 9 we present how the four functional relationships can be explained with combining upstream water area and the local drain direction. The four curves can be separated in two groups based on the values of upstream water area, then from each group, the left-most curve is the one that corresponds to the off-diagonal values of the local drain direction (ldd) encoding. This relatively simple distinction can be understood as a model artifact, because the model uses a square grid. This leads to the cells where the local drain direction is diagonal (45, 135, 225 and 315 degrees) including more channel length. Concretely, these cells should have $\sqrt{2}$ more channel length, and this turns out to roughly be the relation between the right and left curves of each group.

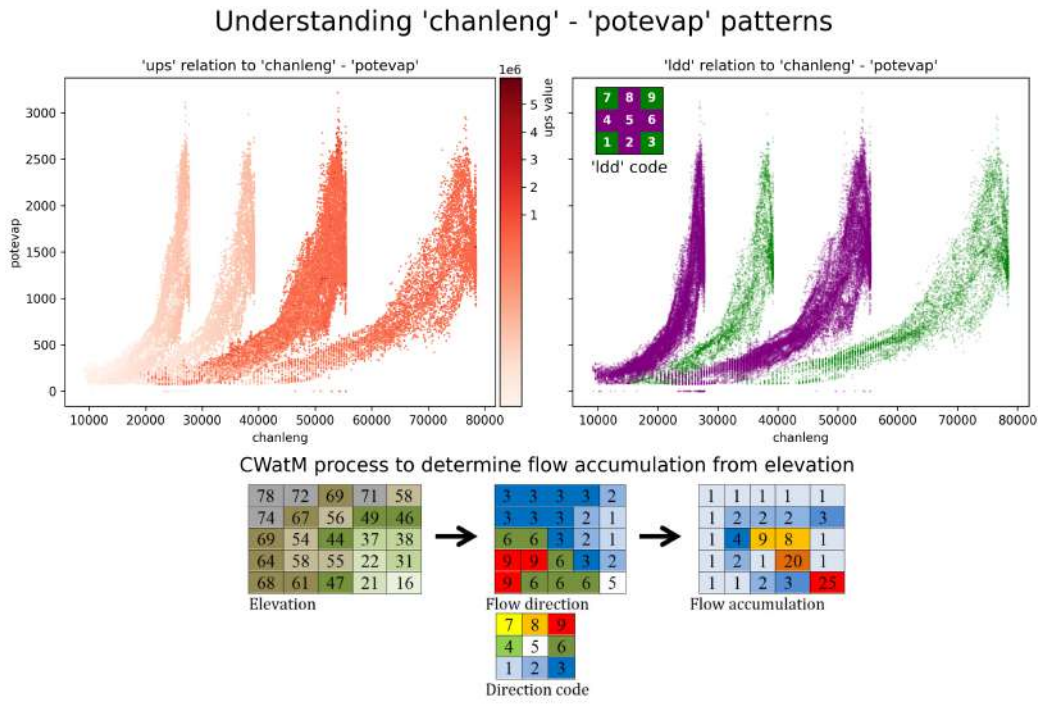


Figure 9: Illustration of how upstream water area (ups) and local drain direction (ldd) explain the four distinct functional relationships between channel length and potential evapotranspiration. This pattern emerges from the model's use of a square grid, where diagonally oriented flow directions effectively increase the measured channel length. Refer to Appendix E.

5 Discussion

5.1 Relevance of the maximal information coefficient

The findings and results of this thesis serve as a confirmation of our initial hypothesis that information theory can be of assistance in exploring the large datasets generated by Global Water Models. We suggest that a combination of traditional and information theory inspired measures of dependence are able to detect and to some extent, characterize complex relationships. By using the Pearson and Spearman correlation coefficient together with the maximal information coefficient (MIC) to explore the relationships between the inputs and outputs of the Community Water Model (CWatM). In doing so, we found that the MIC picked up on different types of relationship that Pearson and Spearman did not, effectively extending the range of relationships that we are able to detect.

The comparison of the ranking these three measures of dependence give to the model inputs (see 5 and A2) show that the MIC detects certain variables which are, from a process understanding perspective, expected to be deemed relevant. Results such as water albedo (albedoWater) ranking the second most important variable for potential evapotranspiration (potevap) or percolation impediment (percolationImp) the most important for ground water recharge (qr), are very reassuring since these are input variables that are expected to be influential for their respective outputs, but which Pearson and Spearman ranked lower.

The method proposed in this work shows the potential of using measures of dependence as a guide in the in-depth exploration of these large dataset. We are able to distinguish simple, threshold-like and complex relationships from each other. While simple relationship are what Pearson and Spearman excel at, threshold relationships have been observed in the previous study by Gnann et al. (2023), and with the method proposed in this work, we propose the MIC as a useful tool to detect them. While in the dataset used in this analysis we have only detected on complex relationship, the one between channel length (chanleng) and potential evapotranspiration (potevap), we strongly believe that this is not an isolated case and that such patterns might occur more than once. Regardless, we believe that the metrics profile we identified (see figure 8) are reasonable and from a methodological perspective a satisfactory confirmation of our initial assumptions.

5.2 Uncovering and understanding unintended model behavior

In this work, we also showcase how permutation feature importance can be applied to a random forest classifier to understand a functional relationship. Specifically, we apply this process to the complex relationship we found between channel length and potential evapotranspiration to understand which are the variables that characterize the different regions we identify. We are able to determine that these relationships appear for two reasons, on one hand we have a distinction between source and down-stream cells and on the other, we uncover a construct of the model caused by the square grid used.

While we consider this example to be of limited hydrological relevance, we believe that the steps followed to reach this conclusion are applicable to the more general task of understanding model behavior. The ad-hoc definition of regions (see figure E1b) and the evaluation of a surrogate model with permutation feature importance can be reproduced for any such relationship. However, depending on the complexity of the underlying process, the interpretation of the results and the determination of the actual source might not be as neat.

5.3 Towards a more holistic evaluation strategy

The results of this work show that the maximal information coefficient is a valuable addition to the set of tools used in the evaluation of Global Water Models. We believe that the application of this methodology to further output variables and models, is a key step towards understanding how these models represent certain processes. With a generic process to understand how the water cycle is represented in GWMs, we believe that applying it to multiple models could shed light into certain unintended consequences or modeling flaws derived from modeling decisions. Such insights into the models' process controls can also be used as feedback to the modeling groups to assess their correctness.

Furthermore, the search for more functional relationships (Gnann et al., 2023) could benefit from the present work by applying this methodology to observational datasets rather than to simulation results. While our results are physically reasonable, this is only a consequence of the model being physically sound. All the conclusions one can draw by evaluating the data generated by the GWMs simulations are limited to the model, and we would advice against seeing them as ground truth. However, if observational data would be used, the findings would relate to the physical system under examination and therefore, one could determine functional relationships that the system exhibits. Such relationships could then be used as constraints of any model that aspires to simulate that system. Note how this is different to calibration and training, since one would be imposing a constraint on the collective of outputs and not on individual instances.

6 Conclusion

In this study, we applied an information theory-based measure of dependence, the maximal information coefficient (MIC), to evaluate and interpret the complex relationships present in data generated by the Community Water Model (CWatM). By comparing MIC to traditional correlation measures, we showed that MIC can identify non-linear and more intricate dependencies that are not captured by Pearson or Spearman correlation coefficients. Our results illustrate that MIC provides valuable insights into the processes governing the water cycle, highlighting variables of physical relevance that remained hidden under conventional analysis approaches.

From a broader perspective, this enhanced capacity to detect and interpret complex relationships contributes to a more comprehensive understanding of Global Water Models (GWMs) and their underlying representations of the water cycle. In doing so, it offers a pathway toward improving model credibility, ultimately benefiting policy makers and practitioners who rely on robust projections to inform water management strategies and adaptation measures in the face of climate change.

Looking ahead, the principles and methodology presented here can be applied to additional variables, models, and even observational datasets, moving toward a more holistic and robust framework for GWM evaluation. Future research might explore hierarchical implementations of MIC, examine the combination of multiple measures of dependence, and investigate how these insights can be incorporated back into model development. Such endeavors will contribute to refining our collective understanding of the global water cycle (Wagener et al., 2021), enabling more effective adaptation strategies and informed planning.

Appendix A Tables and plots of measures of dependence

Table A1 shows the values and the same ranking structure of the measures of dependence that figure 5 represents visually.

We also provide the equivalent plots (see A2) and tables (see A3) for the measures of dependence computed on the climatic regions proposed in Gnann et al. (2023).

output	measure subcol rank	pearson		spearman		MIC	
		input	value	input	value	input	value
evap-total	1	pr	0.85	pr	0.87	pr	0.61
	2	huss	0.84	interceptcap_grassland	0.83	interceptcap_forest	0.57
	3	interceptcap_forest	0.84	interceptcap_forest	0.83	interceptcap_irrNonPaddy	0.57
	4	interceptcap_irrNonPaddy	0.84	interceptcap_irrNonPaddy	0.83	interceptcap_irrPaddy	0.57
	5	interceptcap_grassland	0.84	interceptcap_irrPaddy	0.83	interceptcap_grassland	0.57
	6	interceptcap_irrPaddy	0.84	huss	0.77	kc_grassland	0.48
potevap	1	tasmax	0.91	tasmax	0.94	tasmax	0.86
	2	rsds	0.91	tas	0.92	albedoWater	0.81
	3	tas	0.89	rsds	0.91	tas	0.81
	4	tasmin	0.86	tasmin	0.89	rsds	0.81
	5	rlds	0.77	rlds	0.82	tasmin	0.76
	6	cellarea30min	0.73	cellarea30min	0.79	chanleng	0.76
tw	1	albedoLand	0.46	pr	0.66	pr	0.42
	2	lambda1	0.33	firstStorDepth	0.54	percolationImp	0.29
	3	lambda2	0.31	secondStorDepth	0.54	secondStorDepth	0.29
	4	sfcwind	0.3	chanbw	0.52	firstStorDepth	0.29
	5	lambda3	0.27	hurs	0.47	chanbw	0.29
	6	efficiency	0.24	interceptcap_irrNonPaddy	0.43	lambda3	0.29
qtot	1	pr	0.9	pr	0.85	pr	0.53
	2	interceptcap_forest	0.51	chanbw	0.68	chanbw	0.36
	3	interceptcap_irrNonPaddy	0.51	hurs	0.67	hurs	0.34
	4	interceptcap_irrPaddy	0.51	chanbnkf	0.49	minSoilDepthFrac_grassland	0.25
	5	interceptcap_grassland	0.51	interceptcap_grassland	0.48	minSoilDepthFrac_irrNonPaddy	0.25
	6	huss	0.48	interceptcap_forest	0.48	minSoilDepthFrac_irrPaddy	0.25
qr	1	pr	0.8	pr	0.68	percolationImp	0.84
	2	interceptcap_forest	0.55	firstStorDepth	0.66	secondStorDepth	0.76
	3	interceptcap_irrNonPaddy	0.55	secondStorDepth	0.66	firstStorDepth	0.76
	4	interceptcap_irrPaddy	0.55	interceptcap_grassland	0.58	pr	0.47
	5	interceptcap_grassland	0.55	interceptcap_irrPaddy	0.58	minSoilDepthFrac_grassland	0.42
	6	huss	0.52	interceptcap_irrNonPaddy	0.58	minSoilDepthFrac_forest	0.42

Figure A1: Ranking of the top 6 most controlling inputs for 5 selected outputs according to Pearson, Spearman and MIC. These measures are computed on the global dataset.

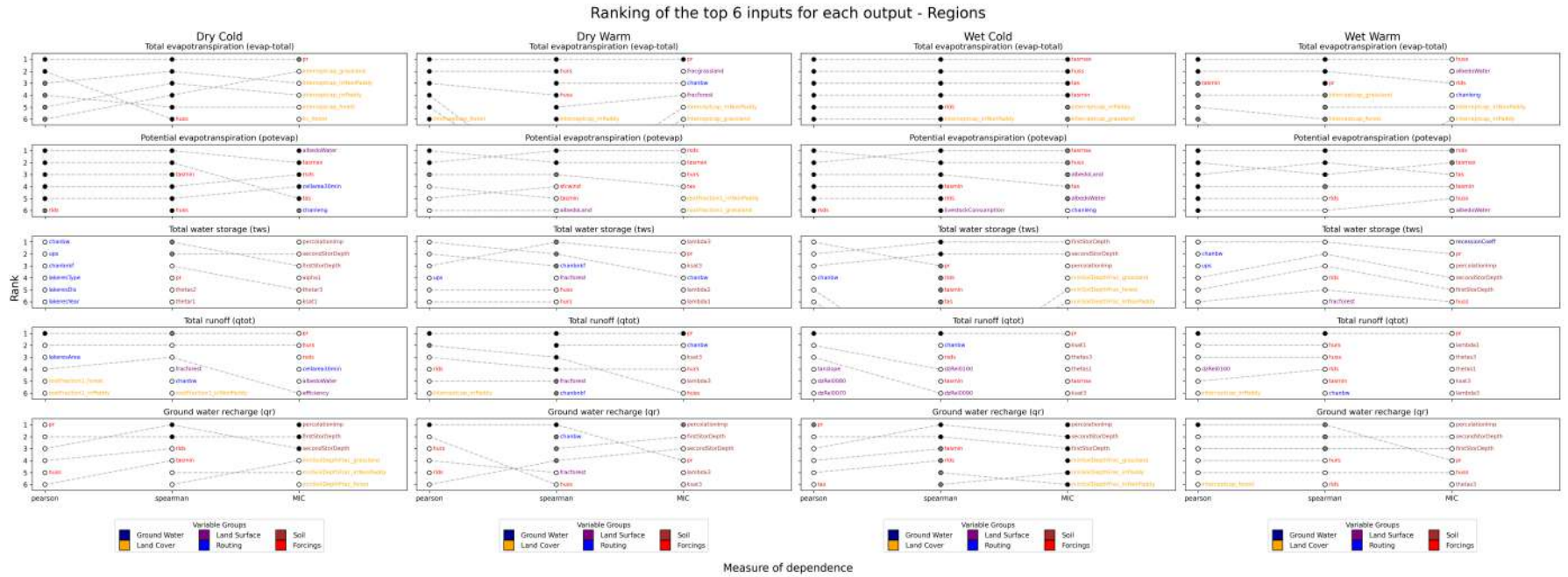


Figure A2: Ranking of the top 6 most controlling inputs for 5 selected outputs according to Pearson, Spearman and MIC. These measures are computed on four different climate regions (see Appendix B). The input variables are grouped and colored according to their domain. To indicated the actual value of the measures of dependence, the dots are colored black for value above 0.7, gray for values between 0.6 and 0.7 and white for those below 0.6. A table with numerical values is provided in A3.

	region	dry cold						dry warm						wet cold						wet warm						
		pearson		spearman		MIC		pearson		spearman		MIC		pearson		spearman		MIC		pearson		spearman		MIC		
		subcol	input	value	input	value	input	value	input	value	input	value	input	value	input	value	input	value	input	value	input	value	input	value		
		output	rank																							
evap-total		1	pr	0.88	pr	0.88	pr	0.85	pr	0.92	pr	0.95	pr	0.8	tasmax	0.91	tasmax	0.93	tasmax	0.77	huss	0.78	huss	0.81	huss	0.56
		2	huss	0.72	interceptcap_inNonPaddy	0.73	interceptcap_grassland	0.49	huss	0.77	huss	0.79	fracgrassland	0.55	huss	0.9	huss	0.92	huss	0.75	rids	0.75	rids	0.78	albedoWater	0.53
		3	interceptcap_inNonPaddy	0.62	interceptcap_inPaddy	0.73	interceptcap_inNonPaddy	0.49	huss	0.76	charneng	0.78	charneng	0.54	tas	0.89	tas	0.91	tas	0.74	tasmin	0.69	pr	0.7	rids	0.6
		4	interceptcap_forest	0.62	interceptcap_grassland	0.73	interceptcap_inPaddy	0.49	interceptcap_grassland	0.72	huss	0.75	fracforest	0.52	tasmin	0.87	tasmin	0.89	tasmin	0.7	interceptcap_grassland	0.69	interceptcap_grassland	0.69	charneng	0.47
		5	interceptcap_inPaddy	0.62	interceptcap_forest	0.73	interceptcap_forest	0.49	interceptcap_inNonPaddy	0.72	fracforest	0.75	interceptcap_inNonPaddy	0.51	rids	0.85	rids	0.88	interceptcap_inPaddy	0.68	interceptcap_forest	0.69	interceptcap_inNonPaddy	0.69	interceptcap_inNonPaddy	0.43
		6	interceptcap_grassland	0.62	huss	0.71	kc_forest	0.45	interceptcap_forest	0.72	interceptcap_inPaddy	0.71	interceptcap_grassland	0.51	interceptcap_inNonPaddy	0.82	interceptcap_inNonPaddy	0.84	interceptcap_grassland	0.68	interceptcap_inPaddy	0.69	interceptcap_forest	0.69	interceptcap_inPaddy	0.43
pctevap		1	tasmax	0.92	tasmax	0.95	albedoWater	0.77	tasmax	0.75	rids	0.74	rids	0.45	huss	0.91	tasmax	0.92	tasmax	0.68	rids	0.91	rids	0.9	rids	0.65
		2	tas	0.89	tas	0.92	tasmax	0.76	rids	0.74	tasmax	0.74	tasmax	0.4	tasmax	0.89	huss	0.92	huss	0.67	tasmax	0.86	tasmax	0.74	tasmax	0.62
		3	tasmin	0.84	tasmin	0.88	rids	0.72	tas	0.88	tas	0.88	huss	0.37	tas	0.85	tas	0.89	albedoLand	0.65	tas	0.86	tasmax	0.69	tas	0.58
		4	rids	0.79	rids	0.87	cellarea30min	0.7	tasmin	0.52	stowind	0.56	tas	0.35	tasmin	0.83	tasmin	0.87	tas	0.63	tasmin	0.82	tasmin	0.68	tasmin	0.56
		5	cellarea30min	0.77	cellarea30min	0.86	tas	0.7	stowind	0.48	tasmin	0.5	roofFraction1_inNonPaddy	0.52	rids	0.8	rids	0.84	albedoWater	0.6	rids	0.77	rids	0.59	huss	0.51
		6	rids	0.65	huss	0.76	charneng	0.68	albedoLand	0.41	albedoLand	0.43	roofFraction1_grassland	0.52	rids	0.79	livestockConsumption	0.79	charneng	0.59	huss	0.75	huss	0.57	albedoWater	0.51
hwa		1	charneng	0.23	firstStorDepth	0.6	percolationImp	0.36	charneng	0.38	pr	0.69	lambda3	0.4	pr	0.26	firstStorDepth	0.72	firstStorDepth	0.54	pr	0.48	pr	0.52	recessionCoeff	0.38
		2	ups	0.21	secondStorDepth	0.6	secondStorDepth	0.35	charneng	0.33	charneng	0.67	pr	0.4	firstStorDepth	0.14	secondStorDepth	0.72	secondStorDepth	0.54	charneng	0.45	secondStorDepth	0.42	pr	0.24
		3	charneng	0.17	theta3	0.5	firstStorDepth	0.35	pr	0.32	charneng	0.6	ksat3	0.38	secondStorDepth	0.14	pr	0.65	percolationImp	0.53	ups	0.37	firstStorDepth	0.42	percolationImp	0.24
		4	lakeareaType	0.14	pr	0.49	alpha1	0.32	ups	0.29	fracforest	0.59	charneng	0.38	charneng	0.12	rids	0.63	minSoilDepthFrac_grassland	0.45	secondStorDepth	0.36	rids	0.42	secondStorDepth	0.23
		5	lakeareaDis	0.14	theta2	0.47	theta3	0.32	huss	0.27	huss	0.59	lambdas2	0.36	minSoilDepthFrac_inNonPaddy	0.11	tasmin	0.63	minSoilDepthFrac_forest	0.45	firstStorDepth	0.36	huss	0.42	firstStorDepth	0.23
		6	lakeareaYear	0.12	theta1	0.45	ksat1	0.31	huss	0.23	huss	0.57	lambdas1	0.36	minSoilDepthFrac_forest	0.11	tas	0.62	minSoilDepthFrac_inNonPaddy	0.45	huss	0.33	fracforest	0.42	huss	0.21
qtot		1	pr	0.71	pr	0.67	pr	0.92	pr	0.85	pr	0.91	pr	0.72	pr	0.88	pr	0.7	pr	0.44	pr	0.9	pr	0.84	pr	0.55
		2	huss	0.5	huss	0.52	huss	0.29	huss	0.63	charneng	0.74	charneng	0.51	dzRe0100	0.45	charneng	0.38	ksat1	0.2	huss	0.41	huss	0.41	lambdas1	0.22
		3	lakearea	0.3	efficiency	0.39	rids	0.29	huss	0.58	huss	0.71	ksat3	0.42	dzRe0090	0.43	rids	0.33	theta3	0.2	huss	0.27	huss	0.38	theta3	0.21
		4	efficiency	0.28	fracforest	0.37	cellarea30min	0.26	rids	0.48	huss	0.7	huss	0.42	tasmin	0.42	dzRe0100	0.43	theta1	0.19	dzRe0100	0.25	rids	0.36	theta1	0.2
		5	roofFraction1_forest	0.28	charneng	0.36	albedoWater	0.24	fracforest	0.48	fracforest	0.64	lambdas3	0.41	dzRe0080	0.41	tasmin	0.28	tasmin	0.19	rids	0.24	tasmin	0.34	ksat3	0.2
		6	roofFraction1_inPaddy	0.28	roofFraction1_inNonPaddy	0.3	efficiency	0.23	interceptcap_inPaddy	0.46	charneng	0.62	huss	0.4	dzRe0070	0.4	dzRe0090	0.27	ksat3	0.19	interceptcap_inPaddy	0.24	charneng	0.31	lambdas3	0.19
qr		1	pr	0.42	secondStorDepth	0.79	percolationImp	0.97	pr	0.7	pr	0.72	percolationImp	0.86	pr	0.65	secondStorDepth	0.79	percolationImp	0.98	pr	0.72	pr	0.66	percolationImp	0.59
		2	firstStorDepth	0.31	firstStorDepth	0.79	firstStorDepth	0.91	huss	0.53	charneng	0.61	firstStorDepth	0.54	firstStorDepth	0.47	firstStorDepth	0.79	secondStorDepth	0.95	secondStorDepth	0.53	secondStorDepth	0.62	secondStorDepth	0.37
		3	secondStorDepth	0.31	rids	0.57	secondStorDepth	0.9	huss	0.48	firstStorDepth	0.6	secondStorDepth	0.54	secondStorDepth	0.47	tasmin	0.67	firstStorDepth	0.95	firstStorDepth	0.53	firstStorDepth	0.62	firstStorDepth	0.37
		4	rids	0.29	tasmin	0.55	minSoilDepthFrac_grassland	0.55	fracforest	0.45	secondStorDepth	0.6	pr	0.52	tasmin	0.43	rids	0.67	minSoilDepthFrac_grassland	0.79	huss	0.43	huss	0.47	huss	0.24
		5	huss	0.29	minSoilDepthFrac_inNonPaddy	0.53	minSoilDepthFrac_inNonPaddy	0.55	rids	0.42	fracforest	0.59	lambdas3	0.39	rids	0.42	minSoilDepthFrac_inNonPaddy	0.67	minSoilDepthFrac_inPaddy	0.79	huss	0.36	huss	0.45	huss	0.24
		6	tasmin	0.26	minSoilDepthFrac_grassland	0.53	minSoilDepthFrac_forest	0.55	secondStorDepth	0.41	huss	0.58	ksat3	0.38	tas	0.41	minSoilDepthFrac_inPaddy	0.67	minSoilDepthFrac_inNonPaddy	0.79	interceptcap_forest	0.33	rids	0.43	theta3	0.2

Figure A3: Ranking of the top 6 most controlling inputs for 5 selected outputs according to Pearson, Spearman and MIC. These measures are computed on four different climate regions (see Appendix B).

Appendix B Climate regions defined in previous work

For reference, figure B1 shows the climate regions defined in Gnann et al. (2023). These climate regions are classified as wet or dry based on the aridity index (N/P) being below or above 1, respectively, and as cold or warm depending on how many days per year the temperature stays below 1°C .

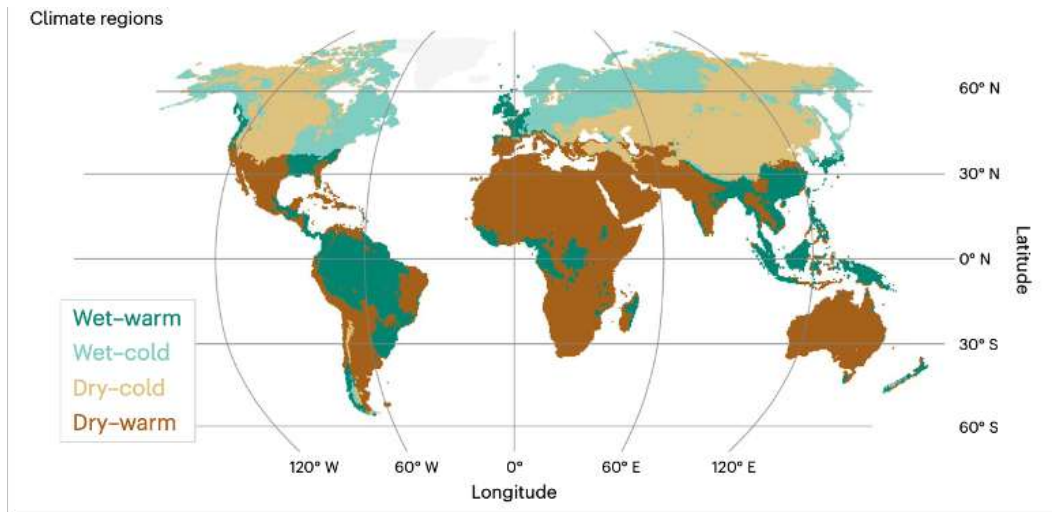


Figure B1: Climate regions defined according to aridity and temperature. Figure from the original paper by Gnan et al. (2023).

Appendix C Statistical significance of the MIC values

When computing a measure of dependence, it is critical to ensure that the obtained values are statistically significant and not a consequence of an abnormal data distribution. To this end, we created 25 shuffled datasets on which we computed all the measures of dependence. These data can be found in the folder `data/processed/bivariate_metrics/CWatM/shuffled` of the thesis repository.

We then use these to test our null hypothesis of independence. However, since the number of variable pairs is large, it is important to control the false discovery rate. We do so following the Benjamini and Hochberg procedure (Benjamini & Hochberg, 1995) as suggested by the original authors in Reshef et al. (2011).

All values of MIC provided throughout this work are deemed significant at a probability of false rejection $\alpha = 0.05$. Figure C1 shows the Benjamini and Hochberg procedure applied to the climatic region "wet warm". We have selected this example over that of the global dataset because this region presents large p-values which provide a better understanding of the task at hand. The equivalent plots for the global data and the rest of the climate regions can be found in the thesis repository under `reports/figures/CWatM.data/plot_benjamini_hochberg*.png`.

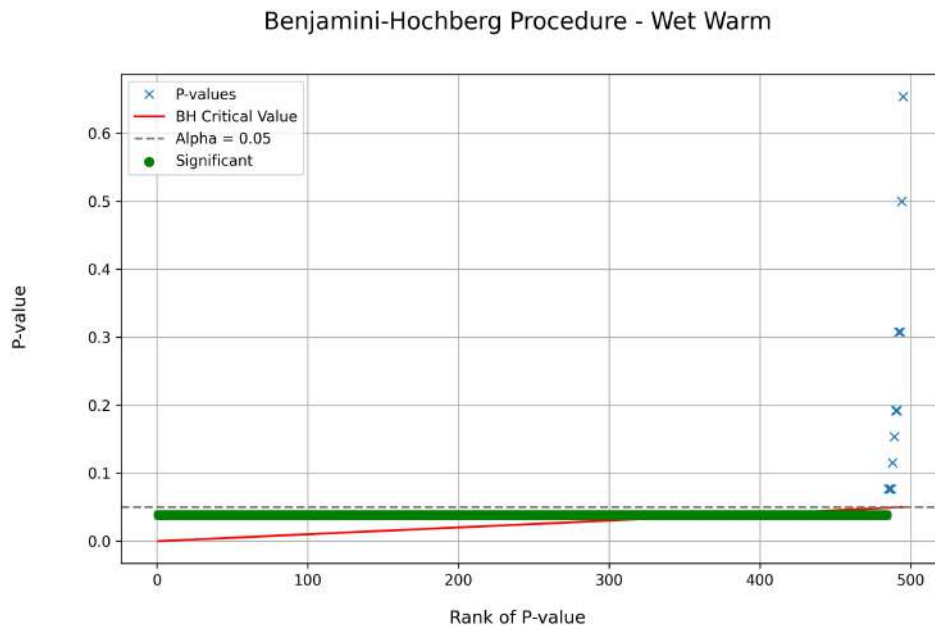


Figure C1: Visualization of the Benjamini and Hochberg procedure applied to the Wet Warm region. The plot shows the p-values of all the variable pairs being tested with crosses. The red line depicts the critical value of the Benjamini and Hochberg procedure. The variable pairs that are deemed significant are marked with green dots.

Appendix D Metric profiles for each output

In figure D1 we provide the bar plots of the metrics profiles for the top 6 inputs (by MIC value) for each output. This is yet another graphical representation of the data presented in the table A1. This visualization proved to be helpful in identifying the metric profiles presented in figure 8.

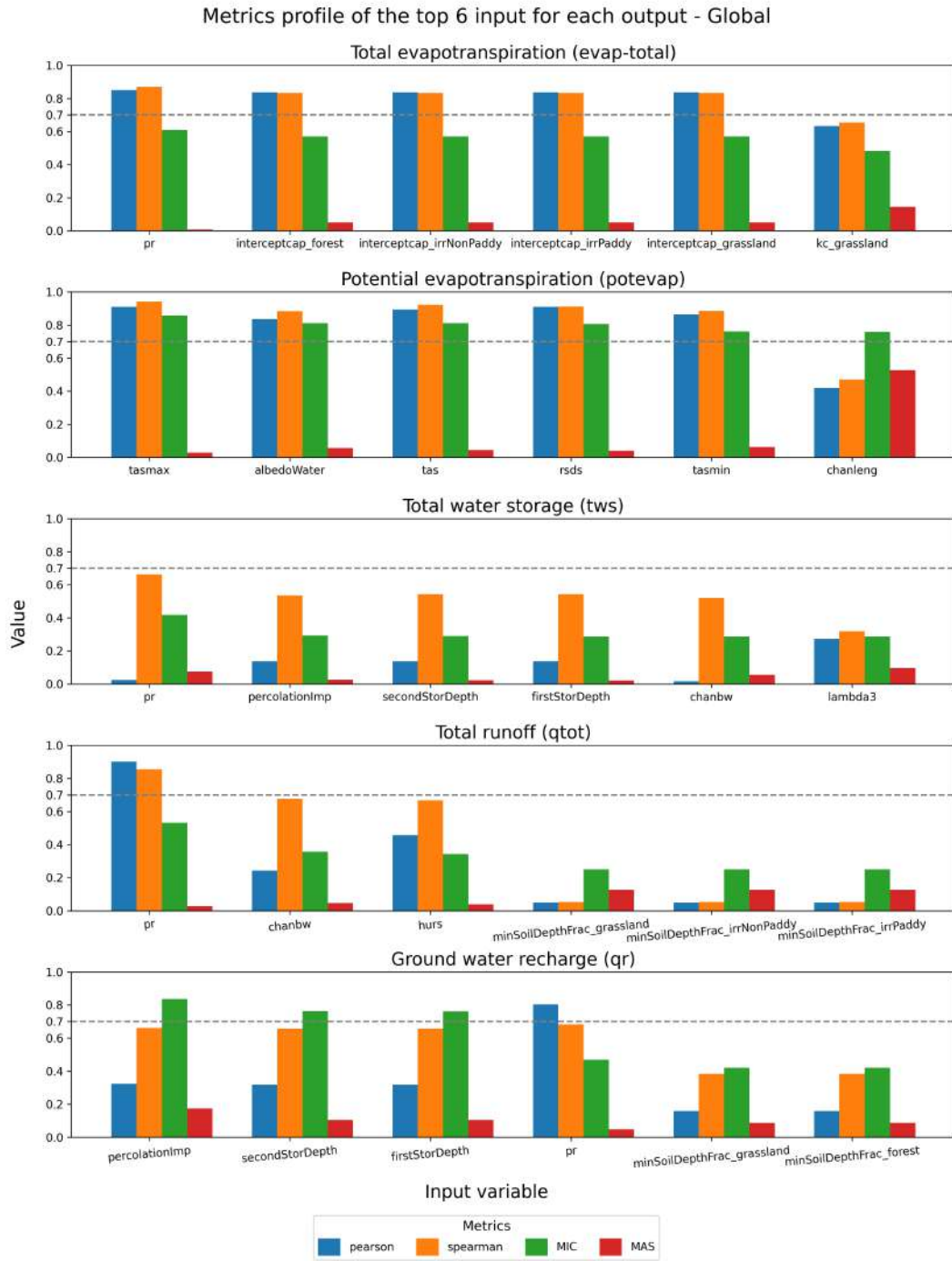


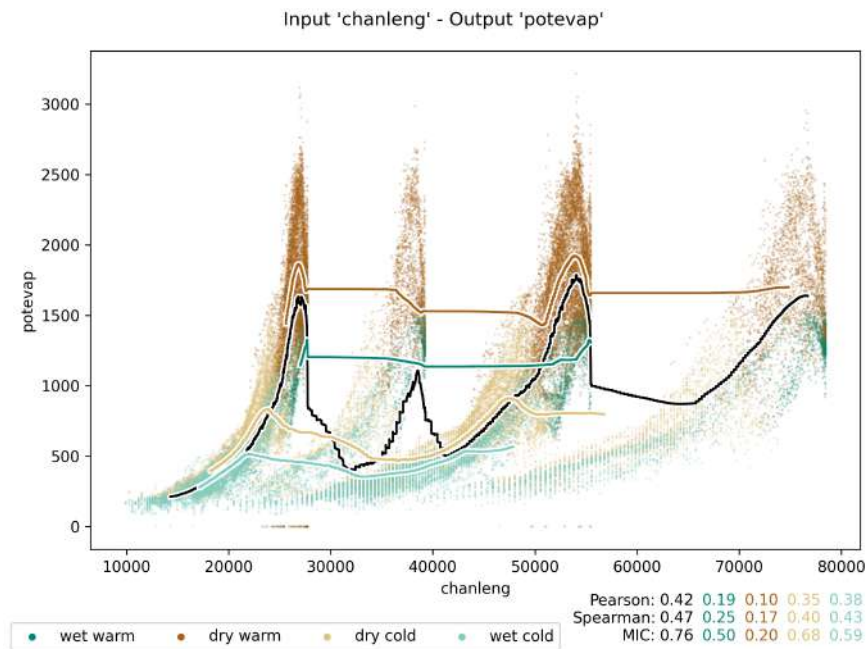
Figure D1: Metric profiles for the top 6 inputs (by MIC value) for each output.

Appendix E Understanding patterns with a Random Forest

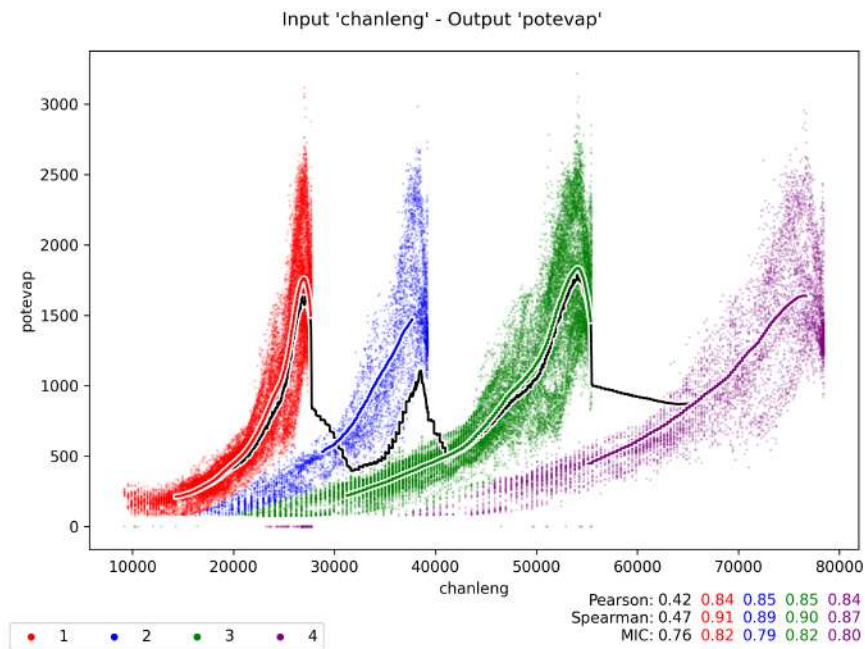
Upon observing the complex relationship found between channel length (chanlength) and potential evapotranspiration (potevap), we set off to understand them. Initially we thought they might be defined by the climate regions suggested in Gnann et al. (2023) and shown in figure B1. However, the scatterplot E1a shows clearly how this regionalization does not yield an informative result. To further study and understand these relationships we decided to use a random forest classifier. These models, as argued in section 2.2, provide more robust predictions than a single decision trees by design.

To train such a classifier, we first had to create the training data. In this case we did so ad-hoc and we created the four regions shown in E1b. We then removed the channel length variable from the features and we fitted a "scikit-learn" `RandomForestClassifier` with the target being the four regions we manually created. The model was evaluated using 5-fold cross validation yielding a Area Under the Receiver Operating Characteristic Curve (ROC AUC) score of 0.984. The ROC AUC score was used in a one-vs-rest set up since we are dealing with a multi-class scenario.

We then trained a classifier on a stratified 80% of the data, and we computed the permutation feature importance on the remaining 20% with the `sklearn.inspection.permutation_importance` function and with 20 repetitions. The results presented in figure E2 show that the upstream water area (ups), the local drain direction (ldd) and the channel Manning index (chanman) are the most important variables.



(a) Potential evapotranspiration vs. channel length with Gnann regions.



(b) Potential evapotranspiration vs. channel length with manually labeled regions.

Figure E1: Scatterplots of potential evapotranspiration against channel length for two different regionalizations. Data is colored according to the regions defined by Gnann (see Appendix B) and according to the manually labeled ones. Black corresponds to the joint global dataset. The lines show the average relationship per region. In the bottom right corner, the values of the measures of dependence are shown.

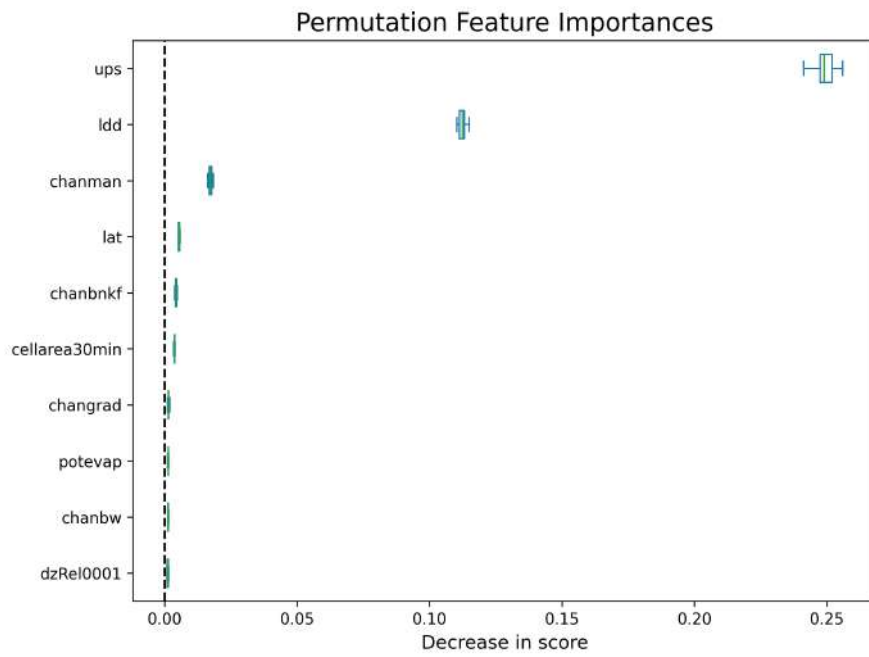


Figure E2: Results of the permutation feature importance shows the decrease in score when a given variable is randomly shuffled. Each feature was permuted 20 times to provide an estimation of uncertainty, the narrow box plots show that the results are very consistent.

Open Research Section

The data used in this study can be found in the GitHub repository https://github.com/maxserra/master_thesis. The code, scripts and notebooks used to process the data are likewise found in this repository. The raw data obtained from the CWatM modeling group is not provided, but the products of it's processing are.

Acknowledgments

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